

Gijae Ahn

Robotics Engineer

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Education

Pusan National University (PNU) , Electrical & Electronic Engineering	Busan, Korea
• Thesis: Two Studies in Robotic Manipulation: Development of a Lightweight Compliant 6-DoF Arm and Dynamic Non-Prehensile Object Transfer	Mar 2024 – Feb 2026
Pusan National University (PNU) , Electrical Engineering	Busan, Korea
• Military Service: Oct 2018	Mar 2018 – Feb 2024
• Jun 2020	

Competitions

- RoboCup@Home 2025 (Salvador) - First Place: Led development of a lightweight compliant 6-DoF manipulator (HW/SW); contributed controller, motion planning, and high-level control API.
- RoboCup@Home 2024 (Eindhoven) - Second Place: Developed arm controller, motion planning modules, and high-level control API interface.
- RoboCup@Home 2023 (Bordeaux) - First Place: Hardware engineering contributions.
- ZEUS Industrial Robot ZERO Mission Challenge 2024 - First Place (Presidential Award, Republic of Korea): Team lead for Robot Bartender; implemented arm controller and motion planning; integrated an LLM-based interaction pipeline.
- Space Exploration & Development Robot Challenge 2023 (Korea) - First Place (Prime Minister's Award, Republic of Korea): Developed a 4-DoF rover arm hardware and controller for a lunar-environment mock mission.
- Baemin Food Delivery Challenge (Real Robot) 2023 - Second Place (President's Award, Korea Institute for Robotics Industry Advancement, KIRIA): Developed a low-level motor driver interface for the real-robot delivery platform.
- Baemin Autonomous Delivery Challenge (Simulation) 2023 - First Place (Ministerial Award, Ministry of Trade, Industry and Energy, MOTIE): Contributed to simulation task setup and overall performance.

Publications

Dynamic Scoop-and-Flick Manipulation for Rapid Non-Prehensile High-Arc Object Transfer	2026
G. Ahn, J. Lee, S. Oh, M. Shin, S.-J. Yi, J. Seo (ICRA 2026 (Accepted))	
ANUBIS: A Compact, Low-Cost, Compliant Humanoid Mobile Manipulation Robot	2025
T. Kang, J. Kim, S. Nasrat, D. Song, G. Ahn, M. Jo, S. Lee, S.-J. Yi (IEEE-RAS Humanoids 2025)	
Lucio at RoboCup@Home: An Open-Hardware Mobile Manipulator with Modular Software and On-Device LLM Planning	2025
D. Song, T. Kang, S. Nasrat, J. Kim, M. Jo, G. Ahn, S. Lee, S. Yi (IEEE-RAS Humanoids 2025 (Poster Presentation))	
Development of Dual-Arm Human Companion Robots That Can Dance	2024
J. Kim, T. Kang, D. Song, G. Ahn, S.-J. Yi (Sensors, 24(20), 6704)	

Experience

Tomorro Robotics Korea, Research Intern

Korea

CES 2026 demo system for autonomous factory-line collaboration.

Sept 2025 – present

9 months

- Led engineering for an autonomous mobile manipulator in a dual-robot workflow.
- Implemented end-to-end pipeline for mobile base detection, grasping, and box transport to a conveyor.
- Ensured synchronization with a secondary VLA-based packing robot.

Projects

Autonomous Quadruped Patrol System

Government and National Police Agency related project.

- Built perception coverage for safe navigation using four Intel RealSense D435 cameras.
- Owned camera mount design and a multi-camera depth pipeline (stitching/fusion) to generate obstacle maps.
- Supported collision-free locomotion on patrol routes with anomaly detection (suspects/illegal parking).

Mobile Dual-Arm Robot System for Scalable Learning Data Collection

LG Electronics industry-academia project.

- Developed a mobile bimanual manipulation platform optimized for data collection and safe policy rollouts.
- Led development of a lightweight compliant manipulator to reduce environment-damage risk while controlling cost.
- Extended the platform into the M.S. thesis direction.

Skills

Technical Areas: Kinematics & Dynamics, Sensor Fusion, Control, Motion Planning, Optimization, Object Recognition, Deep Learning

Languages: C++, Python, Matlab

Frameworks: ROS / ROS2, MuJoCo, Webots, Isaac Gym, Drake

Software Tools: CAD(Onshape), Git, Docker, PyTorch, OpenCV, CMake, Make

Team-based Development: Git-based collaboration (branching, code reviews) and collaborative CAD part design and assembly in Onshape for integrated robot hardware.